

Results — proposed reorganisation (v4 draft)

Environmental Provisions in Chinese PTAs and Export Composition

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What this document is. A standalone draft of a reorganised Section 5. Six subsections in place of twelve, ordered: full panel → collapsed panel and inference → control groups → dynamics → sub-indices → extensive margin. All figures come from the corrected table document; the numbers that changed relative to `paper_v3` are flagged in footnotes. Cross-references to appendix tables are placeholders. Nothing in `paper_v3.tex` has been modified.

Specification recap. The estimating equation is

$$\ln x_{fptd} = \beta_1 EP_{dt} \times g_p + \beta_2 EP_{dt} \times b_p + \gamma_1 TD_{dt} \times g_p + \gamma_2 TD_{dt} \times b_p + \theta_{fptd} + \theta_{fpt} + \varepsilon_{fptd}, \quad (1)$$

with g_p and b_p the green and pollution-intensive product indicators, EP_{dt} environmental depth and TD_{dt} non-environmental agreement depth. Standard errors are clustered by destination throughout; because the design has 23 treated destinations, the wild cluster bootstrap ($B = 9,999$) rather than the asymptotic p -value is the reference inference.

1 Results

1.1 The full panel

Table 1 reports the reference estimates. Equation (1) is estimated on the full firm–product–destination–year panel: 21.5 million observations, 225 destination clusters, firm $\times$ product $\times$ destination, firm $\times$ destination $\times$ year and product $\times$ year fixed effects. Columns (1)–(3) use the World Bank EP depth index, columns (4)–(6) the TREND provision count; within each block the outcome is log export value, log quantity and log unit value. Panel A controls for non-environmental World Bank depth, Panel B for the independently coded DESTA index (DUR2014).

The table is deliberately wide, because the central claim of this section is a claim about *invariance*: the green coefficient does not move across any of these dimensions, and the dirty coefficient moves across exactly one of them.

The green margin. Zero everywhere. On export value the WB estimate is -0.0023 (s.e. 0.0039, bootstrap $p = 0.69$) under the World Bank depth control and -0.0019 (0.0037, $p = 0.67$) under DESTA; the TREND estimate is -0.0001 in both, with bootstrap p -values of 0.93 and 0.92. The joint F -test on all four composition interactions gives $p = 0.31$ (WB) and $p = 0.71$ (TREND) in Panel A, and $p = 0.18$ and $p = 0.26$ in Panel B. Quantity and unit value behave identically: across the eight green cells in columns (2), (3), (5) and (6), no bootstrap p -value falls below 0.74.

That last point does work that the value regression alone cannot do. A null on export value is consistent with two offsetting responses — more units at lower prices, or fewer units at higher prices

— cancelling in the aggregate. Columns (2) and (3) rule this out directly: neither the volume channel nor the price channel moves. The green null is a null on both margins separately, not a coincidence of the value aggregate.¹

The dirty margin. Negative throughout, and this is the one place in the paper where a modelling choice changes the verdict. Under the World Bank depth control the WB×dirty coefficient is -0.0044 with asymptotic $p = 0.052$, but the bootstrap raises it to $p = 0.19$; the TREND counterpart is -0.0009 (0.15 asymptotic, 0.18 bootstrap). Under the DESTA control the same coefficients are -0.0056 (0.039 asymptotic, 0.049 **bootstrap**) and -0.0014 (0.048 asymptotic, 0.069 bootstrap). Same data, same fixed effects, same estimation sample; the only difference is which measure of non-environmental agreement depth is partialled out.²

The mechanical explanation is collinearity. Net of the fixed effects, environmental depth correlates 0.959 with World Bank non-environmental depth and 0.891 with DESTA depth. The more the control overlaps the regressor of interest, the less independent variation is left and the wider the standard error. Read that way, Panel B is the better-identified specification and Panel A is over-controlled — which would make the dirty effect real and the World Bank specification the one hiding it.

We do not adopt that reading, and Section 1.2 explains why: the permutation test, which reassigns entire EP profiles across treated destinations and is therefore indifferent to which depth control is in the regression, assigns this coefficient a p -value between 0.15 and 0.38 in every one of the four sample-control variants. But the ambiguity is real and we flag it here rather than resolving it by choosing the control that gives the answer we prefer.

Quantity, not price. Where the dirty coefficient does move, it moves on quantity. Under the DESTA control it is -0.0070 on log quantity (asymptotic $p = 0.020$, bootstrap $p = 0.035$) — larger in magnitude and more precisely estimated than on value — while the unit value coefficient is $+0.0004$ with $p = 0.68$. Taking Tables 1 and 3 together, sixteen cells estimate an EP interaction on log unit value; exactly one is nominally significant (TREND×dirty in the collapsed panel, asymptotic $p = 0.076$) and it does not survive the bootstrap ($p = 0.17$). Whatever this coefficient reflects, it is a shipment-volume margin with no price response. That is consistent with a compositional shift in what firms ship and inconsistent with a quality-upgrading or cost-passthrough interpretation.

How large a null. A null result means one of two things: the effect is absent, or the design cannot see it. The bootstrap confidence intervals say which. On the full panel the 95% bootstrap interval for EP×green is $[-0.0353, +0.0355]$ per World Bank provision — about $\pm 3.5\%$, or, since one standard deviation of WB EP depth in the estimation sample is roughly 2.4 provisions, at most a $\pm 8\text{--}9\%$ change in green relative to neutral exports. The corresponding asymptotic interval, $[-0.0100, +0.0055]$, is roughly six times narrower; reporting that width would claim a precision this paper’s own inference section declares unwarranted.

Table 2 places the bound against the closest benchmark. BRANDI2020 find that one liberal environmental provision raises the green share of developing-country exports by around 0.4 percentage points, which is roughly $+0.16$ log points in the metric of equation (1). The upper end of the interval here is about one quarter of that figure. The design can therefore exclude effects at or above the

¹This replaces the summary table in `paper_v3` §5.10, which reported ranges of p -values and noted that exact coefficients were “available upon request”. They are now in columns (2), (3), (5) and (6).

²In `paper_v3` this contrast is invisible, because the tables reported only the World Bank depth control in the main text and the sentence “the TREND index never shows the effect” was written from the collapsed-panel figure. On the full panel with the DESTA control the TREND index does show it. The corrected tables make the contrast the substantive question it is.

Brandi magnitude; it cannot exclude smaller positive effects. Appendix Table 1.6 reports the implied minimum detectable effects in per-provision and per-standard-deviation terms.

Table 1: Full panel — triple-difference estimates

	WB EP depth (EP_{dt}^{WB})			TREND EP count (EP_{dt}^{TR})		
	(1) $\ln x_{fpdt}^{val}$	(2) $\ln x_{fpdt}^{qua}$	(3) $\ln x_{fpdt}^{uv}$	(4) $\ln x_{fpdt}^{val}$	(5) $\ln x_{fpdt}^{qua}$	(6) $\ln x_{fpdt}^{uv}$
<i>Panel A — Non-environmental depth control (TD_{dt}, World Bank)</i>						
$EP_{dt} \times g_p$	−0.0023 (0.0039)	−0.0013 (0.0041)	−0.0004 (0.0013)	−0.0001 (0.0010)	−0.00003 (0.0011)	−0.0001 (0.0003)
Asymptotic p	[0.566]	[0.750]	[0.748]	[0.924]	[0.974]	[0.857]
Bootstrap p	[0.686]	[0.851]	[0.740]	[0.931]	[0.978]	[0.860]
$EP_{dt} \times b_p$	−0.0044* (0.0022)	−0.0048* (0.0029)	0.0001 (0.0010)	−0.0009 (0.0006)	−0.0006 (0.0007)	−0.0001 (0.0004)
Asymptotic p	[0.052]	[0.096]	[0.903]	[0.153]	[0.409]	[0.697]
Bootstrap p	[0.185]	[0.391]	[0.937]	[0.177]	[0.443]	[0.734]
θ_{fpd}	✓	✓	✓	✓	✓	✓
θ_{fdt}	✓	✓	✓	✓	✓	✓
θ_{pt}	✓	✓	✓	✓	✓	✓
Observations	21,519,511	21,313,047	21,313,037	21,519,511	21,313,047	21,313,037
Clusters (d)	225	225	225	225	225	225
R^2	0.873	0.913	0.963	0.873	0.913	0.963
<i>Panel B — DESTA depth control</i>						
$EP_{dt} \times g_p$	−0.0019 (0.0037)	−0.0015 (0.0034)	−0.0004 (0.0019)	−0.0001 (0.0010)	−0.0001 (0.0010)	−0.0001 (0.0005)
Asymptotic p	[0.613]	[0.660]	[0.819]	[0.904]	[0.917]	[0.776]
Bootstrap p	[0.673]	[0.755]	[0.843]	[0.920]	[0.933]	[0.852]
$EP_{dt} \times b_p$	−0.0056** (0.0027)	−0.0070** (0.0030)	0.0004 (0.0011)	−0.0014** (0.0007)	−0.0014* (0.0008)	−0.00004 (0.0003)
Asymptotic p	[0.039]	[0.020]	[0.676]	[0.048]	[0.080]	[0.903]
Bootstrap p	[0.049]	[0.035]	[0.765]	[0.069]	[0.100]	[0.915]
θ_{fpd}	✓	✓	✓	✓	✓	✓
θ_{fdt}	✓	✓	✓	✓	✓	✓
θ_{pt}	✓	✓	✓	✓	✓	✓
Observations	21,517,666	21,311,206	21,311,196	21,517,666	21,311,206	21,311,196
Clusters (d)	225	225	225	225	225	225
R^2	0.873	0.913	0.963	0.873	0.913	0.963

Notes: Full firm-level panel, excluding Hong Kong and Macao. Columns (1) and (4): log export value ($\ln x_{fpdt}^{val}$). Columns (2) and (5): log export quantity ($\ln x_{fpdt}^{qua}$). Columns (3) and (6): log unit value ($\ln x_{fpdt}^{uv} = \ln x_{fpdt}^{val} - \ln x_{fpdt}^{qua}$). Quantity and unit-value regressions are estimated on the subset of observations with non-missing quantity data. All specifications include $TD_{dt} \times g_p$ and $TD_{dt} \times b_p$ controls (not shown). Panel A uses non-environmental WB depth (TD_{dt}); Panel B uses DESTA depth. Standard errors clustered at the destination level in parentheses. Wild cluster bootstrap ($B = 9,999$) p -values in brackets. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$ (asymptotic).

Source: Chinese Customs Office; World Bank DTA Database; TREND Database.

Table 2: Benchmark against Brandi et al. (2020): equivalent effect and comparison with the confidence intervals

Margin	Brandi effect (log points)	Asy. CI upper	WCB CI upper/coef.	Ratio
Green (liberal)	0.1570	0.0055	0.0355	1/4 (WCB)
Dirty (restrictive)	−0.0513	−0.0044 (point est.)	−	1/12

1.2 The collapsed panel, and what survives robust inference

Table 3 re-estimates equation (1) on HS6–destination–year cells with analytic weights. This is not a robustness check on Table 1. It estimates a different object, and the difference between the two is itself a result.

The firm×destination×year fixed effects in the full panel absorb everything that varies at the firm–destination–year level, so identification comes from reallocation *within* the firm across products. The collapsed panel compares products across all firms shipping to a destination in a year, so any shift in *which* firms export dirty goods enters the coefficient as well. The two estimands differ by exactly the between-firm channel.

On the green margin they agree: -0.0046 collapsed against -0.0023 full panel, both indistinguishable from zero on every inference method. On the dirty margin they differ by a factor of roughly 2.7 (-0.0119 against -0.0044), which decomposes the coefficient: approximately three-fifths of the collapsed dirty estimate, $(0.0119 - 0.0044)/0.0119 = 0.63$, is between-firm composition rather than within-firm reallocation. Whatever moves dirty exports operates mainly through which firms ship, not through what a given firm ships. The full panel therefore remains the reference specification, and the collapsed panel is where the between-firm channel becomes visible.

Why inference lives here. The collapsed panel is also where the design can be stress tested. The permutation test reassigns the EP profile of each agreement — its depth level and its entry years — randomly among the 23 treated destinations and re-estimates the model a thousand times; the treated-only design this requires is not feasible at firm level. The leave-one-out exercise, likewise, is a collapsed-panel exercise. Table 3 therefore carries three rows of inference where Table 1 carries two.

Read down those rows for the WB×dirty coefficient. The asymptotic p -value is below 0.001. The wild cluster bootstrap raises it to 0.072. The permutation test raises it to 0.278: in 27.8% of random reassignments of environmental content across the same 23 destinations, the model produces a coefficient at least as large in absolute value. A coefficient that would have arisen a quarter of the time from randomly relabelling which agreements are environmental is not evidence about environmental content.

The depth-control ambiguity, resolved as far as it can be. Table 4 assembles the dirty coefficient across all four sample–control variants with a single inference method, so the eight cells are directly comparable. Two features stand out. The point estimate is uniformly negative and, within each panel, stable. The bootstrap p -value is not: on the collapsed panel it is 0.072 with the World Bank control and 0.047 with DESTA; on the full panel it is 0.185 and 0.049. Significance tracks the control variable, not the data.

The permutation test is the tiebreaker, because it is the one procedure whose null distribution does not depend on which depth control sits in the regression. It gives $p = 0.278, 0.146, 0.146$ and 0.384 across the four variants — no cell below 0.10. The reading we adopt is therefore not that the dirty coefficient is a false positive, and not that it is an established effect, but that it is a coefficient whose apparent precision is an artefact of how much independent variation a particular depth control leaves behind. On the evidence that does not depend on that choice, it is indistinguishable from a placebo.

Leave-one-out: the mechanism is precision, not leverage. Table 5 re-estimates the dirty coefficient dropping one treated destination at a time. The usual concern — that a result is driven by a single country — is not what the exercise finds. The point estimate sits between -0.0097 and

Table 3: Collapsed panel — triple-difference estimates

	WB EP depth (EP_{dt}^{WB})			TREND EP count (EP_{dt}^{TR})		
	(1) $\ln x_{fpdt}^{val}$	(2) $\ln x_{fpdt}^{qua}$	(3) $\ln x_{fpdt}^{uv}$	(4) $\ln x_{fpdt}^{val}$	(5) $\ln x_{fpdt}^{qua}$	(6) $\ln x_{fpdt}^{uv}$
<i>Panel A — Non-environmental depth control (TD_{dt}, World Bank)</i>						
$EP_{dt} \times g_p$	−0.0046 (0.0069)	−0.0055 (0.0069)	0.0005 (0.0028)	0.0018 (0.0018)	0.0019 (0.0020)	−0.0001 (0.0007)
Asymptotic p	[0.508]	[0.426]	[0.854]	[0.316]	[0.360]	[0.868]
Bootstrap p	[0.649]	[0.525]	[0.851]	[0.390]	[0.433]	[0.867]
Permutation p	[0.597]			[0.160]		
$EP_{dt} \times b_p$	−0.0119*** (0.0029)	−0.0115* (0.0067)	0.0006 (0.0042)	0.0004 (0.0016)	−0.0004 (0.0019)	0.0009 (0.0005)
Asymptotic p	[0.000]	[0.087]	[0.880]	[0.825]	[0.837]	[0.076]
Bootstrap p	[0.072]	[0.250]	[0.947]	[0.857]	[0.886]	[0.169]
Permutation p	[0.278]			[0.817]		
θ_{pd}	✓	✓	✓	✓	✓	✓
θ_{dt}	✓	✓	✓	✓	✓	✓
θ_{pt}	✓	✓	✓	✓	✓	✓
Observations	3,681,023	3,672,709	3,672,709	3,681,023	3,672,709	3,672,709
Clusters (d)	228	228	228	228	228	228
R^2	0.843	0.919	0.966	0.843	0.919	0.966
<i>Panel B — DESTA depth control</i>						
$EP_{dt} \times g_p$	−0.0043 (0.0043)	−0.0084* (0.0044)	0.0038 (0.0042)	0.0016 (0.0020)	0.0010 (0.0022)	0.0006 (0.0008)
Asymptotic p	[0.320]	[0.058]	[0.368]	[0.428]	[0.649]	[0.495]
Bootstrap p	[0.546]	[0.268]	[0.446]	[0.508]	[0.712]	[0.542]
Permutation p	[0.480]			[0.316]		
$EP_{dt} \times b_p$	−0.0113*** (0.0022)	−0.0097** (0.0045)	−0.0012 (0.0029)	−0.0002 (0.0014)	−0.0008 (0.0014)	0.0007 (0.0006)
Asymptotic p	[0.000]	[0.033]	[0.689]	[0.858]	[0.603]	[0.252]
Bootstrap p	[0.047]	[0.311]	[0.757]	[0.905]	[0.876]	[0.332]
Permutation p	[0.146]			[0.904]		
θ_{pd}	✓	✓	✓	✓	✓	✓
θ_{dt}	✓	✓	✓	✓	✓	✓
θ_{pt}	✓	✓	✓	✓	✓	✓
Observations	3,677,333	3,669,025	3,669,025	3,677,333	3,669,025	3,669,025
Clusters (d)	228	228	228	228	228	228
R^2	0.843	0.919	0.966	0.843	0.919	0.966

Notes: Collapsed panel (product×destination×period cells, analytic weights), excluding Hong Kong and Macao. Columns (1) and (4): log export value ($\ln x_{fpdt}^{val}$). Columns (2) and (5): log export quantity ($\ln x_{fpdt}^{qua}$). Columns (3) and (6): log unit value ($\ln x_{fpdt}^{uv}$). All specifications include $TD_{dt} \times g_p$ and $TD_{dt} \times b_p$ controls (not shown). Panel A uses non-environmental WB depth (TD_{dt}); Panel B uses DESTA depth. Standard errors clustered at the destination level in parentheses. Wild cluster bootstrap ($B = 9,999$) p -values in brackets. Permutation p -values (treated-only design, $R = 1,000$) reported for export value only; columns (2)–(3) and (5)–(6) not permuted. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$ (asymptotic).

Source: Chinese Customs Office; World Bank DTA Database; TREND Database.

Table 4: Summary: dirty-product coefficient across four specification variants, with common inference method

	(1)	(2)	(3)	(4)
Collapsed panel: coefficient	-0.01187	-0.01887	-0.01134	-0.00820
Bootstrap p -value	0.072	0.005	0.047	0.195
<i>Full panel</i> : coefficient	-0.00435	-0.00409	-0.00559	-0.00574
Bootstrap p -value	0.185	0.176	0.049	0.035

Notes: Columns: (1) Excl. HK/Macao, TotalDepth control (baseline); (2) Incl. HK/Macao, TotalDepth control; (3) Excl. HK/Macao, DESTA control; (4) Incl. HK/Macao, DESTA control.

Coefficient on the interaction between environmental depth (WB index) and dirty products. This is the only coefficient in the paper that moves: the one on green products is zero everywhere.

All p -values in this table come from the *wild cluster bootstrap*, so all eight cells are comparable.

How to read it. On the *full panel* the result depends on which depth control is used: with the World Bank control the p -value stays above 0.17; with the DESTA control it falls below 0.05. The two controls have different partial correlations with environmental depth net of fixed effects (0.959 and 0.891, respectively): the more a control overlaps with the variable of interest, the less independent variation remains and the larger the standard error.

−0.0133 in all 23 leave-one-out samples; dropping large treated destinations such as India or Pakistan moves it by one to four per cent and leaves the standard error essentially unchanged.

What is unstable is precision. Dropping Australia moves the coefficient by 13%, to −0.0103, but nearly triples its standard error, from 0.0030 to 0.0087, carrying the p -value to 0.24. Dropping South Korea doubles the standard error, giving −0.0097 with $p = 0.09$. Australia and Korea are not outliers exerting leverage on the estimate; they supply the identifying variation, and removing them leaves too little information for a precise estimate rather than revealing a different effect. The coefficient rests on a thin slice of variation concentrated in two destinations — and which two depends on the depth control, since under DESTA the pivotal destination shifts from Australia to Korea. That a modelling choice unrelated to environmental content determines which destination is pivotal is the same fragility diagnosis reached in Table 4 by a different route.

One caveat we do not resolve. Australia and Korea are also among the destinations with the deepest EP coverage in the sample (WB depth 12 and 17). A dose–response reading — the dirty effect is real and concentrated where the dose is largest — is not ruled out by the leave-one-out geometry. The row *High EP depth only* in Table 5, which drops Peru, Switzerland and Korea jointly, is consistent with either interpretation.

Table 5: Leave-one-out: EP×dirty coefficient stability across 23 treated destinations

Excluded destination	WB depth control		DESTA depth control	
	Excl. HK/Macao (1)	Incl. HK/Macao (2)	Excl. HK/Macao (3)	Incl. HK/Macao (4)
<i>None (baseline)</i>	−0.0119***	−0.0189***	−0.0113***	−0.0082*
<i>None, extended dirty</i>	−0.0121***	−0.0191***	−0.0107***	−0.0071
<i>High EP depth only</i>	−0.0271**	−0.0456***	−0.0161	−0.0164
Bangladesh	−0.0118***	−0.0188***	−0.0113***	−0.0082*
Brunei	−0.0119***	−0.0189***	−0.0114***	−0.0082*
Myanmar	−0.0118***	−0.0188***	−0.0112***	−0.0081*
Cambodia	−0.0119***	−0.0189***	−0.0112***	−0.0081*
India	−0.0120***	−0.0190***	−0.0115***	−0.0084**
Indonesia	−0.0119***	−0.0190***	−0.0119***	−0.0087**
Laos, PDR	−0.0119***	−0.0189***	−0.0113***	−0.0082*
Malaysia	−0.0124***	−0.0195***	−0.0110***	−0.0076*
Pakistan	−0.0114***	−0.0183***	−0.0106***	−0.0056
Philippines	−0.0118***	−0.0189***	−0.0115***	−0.0083*
Singapore	−0.0098**	−0.0166***	−0.0128***	−0.0100**
Korea Rep.	−0.0097*	−0.0225**	−0.0104	−0.0085
Sri Lanka	−0.0119***	−0.0189***	−0.0114***	−0.0082*
Thailand	−0.0120***	−0.0189***	−0.0115***	−0.0082*
Vietnam	−0.0122***	−0.0192***	−0.0112***	−0.0080*
Timor-Leste	−0.0119***	−0.0189***		
Iceland	−0.0119***	−0.0189***	−0.0112***	−0.0081*
Switzerland	−0.0133***	−0.0209***	−0.0115***	−0.0084*
Chile	−0.0121***	−0.0191***	−0.0104***	−0.0071
Costa Rica	−0.0114***	−0.0185***	−0.0109***	−0.0076*
Peru	−0.0133***	−0.0204***	−0.0121***	−0.0092**
Australia	−0.0103	−0.0245**	−0.0110***	−0.0048
New Zealand	−0.0119***	−0.0190***	−0.0113***	−0.0081*
Hong Kong		−0.0129***		−0.0112***
Macau		−0.0183***		−0.0082*
Observations	3,681,023	3,769,187	3,677,333	3,759,733
Clusters (<i>d</i>)	228	230	228	230

Notes: OLS on the collapsed panel. Dependent variable: log total exports from China. The coefficient shown is $\hat{\beta}$ on EP×dirty, re-estimated dropping one treated destination at a time. FE: $pd + dt + pt$. Standard errors (not shown) clustered by destination. Columns (1) and (3) exclude Hong Kong and Macao from the sample; columns (2) and (4) include them as additional treated observations. Columns (1)–(2) use WB total depth as the control-depth variable; columns (3)–(4) use DESTA depth. Observations and clusters refer to the baseline specification; individual rows differ slightly due to the dropped destination.

High EP depth only: estimated on the subsample excluding Peru, Switzerland, and Korea jointly — the three destinations with above-median WB EP depth.

Timor-Leste rows are blank in columns (3)–(4): Timor-Leste is absent from the DESTA dataset. Hong Kong and Macau rows are blank in columns (1) and (3): those territories are already excluded in those samples.

Note on inference. Stars are based on asymptotic p -values with destination-clustered SEs. With ≈ 23 treated destinations, these p -values are substantially understated relative to the wild cluster bootstrap: the baseline coefficient in column (1) has asymptotic $p < 0.001$ but bootstrap $p = 0.072$ (Table 1.6). Stars are not comparable to the rest of the paper.

How to read. This is a test of *coefficient stability*, not significance. The relevant question is how much the point estimate moves when one destination is dropped, not how many stars it carries.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$ (asymptotic).

1.3 Control groups and estimation designs

The estimates so far compare green and dirty products against neutral products, in agreement destinations against non-agreement destinations. Both comparisons can be challenged. Neutral products may be poor counterfactuals for green ones if they sit in different parts of the product space; non-agreement destinations may be poor counterfactuals for agreement destinations if China signs agreements with countries on different growth paths. Table 6 replaces each comparison in turn.

C-prod-HS4 restricts attention to HS6 products with a comparable HS4-level match, so a green product is compared with neutral products in the same four-digit heading rather than with the whole neutral universe. *CEM* matches destinations on log GDP per capita, growth and MFN tariff in 2000, reducing the sample to 57 destination clusters that are comparable on pre-treatment observables. *Deep versus shallow* discards untreated destinations altogether: identification comes from variation in EP depth among the 23 PTA partners, so a destination with a shallow environmental chapter is the counterfactual for one with a deep chapter.

Every cell in the table uses the wild cluster bootstrap, so all forty are comparable, and no stars are reported: with this many designs, star-counting invites exactly the selective reading the table is meant to prevent.

The green coefficient stays between -0.0046 and $+0.0005$ in all twenty cells and is never close to conventional significance — the smallest bootstrap p -value anywhere on the green margin is 0.51. This is the result. An artifact of the comparison group would move when the comparison group changes; this does not move when the product comparison is narrowed to within-HS4, when the destination comparison is matched on pre-treatment observables, or when untreated destinations are removed from the design entirely.

The dirty coefficient behaves differently, and in a way that reinforces Section 1.2. It is negative in eight of the ten firm-level cells but never significant once the sample changes: CEM gives -0.0041 ($p = 0.25$) and -0.0043 ($p = 0.12$), deep-versus-shallow gives -0.0034 ($p = 0.50$) and -0.0042 ($p = 0.17$). C-prod-HS4 reverses the sign entirely, to $+0.0066$ and $+0.0033$. A coefficient that survives the removal of the untreated comparison group only in the specification that keeps it is not identifying a robust composition effect.

Further sample variations — adding product–destination–year controls for MFN tariffs, market concentration and antidumping exposure; excluding the eleven ASEAN destinations, which form the largest single cohort; re-including Hong Kong and Macao; and restricting to products with common support across agreement and non-agreement destinations — are reported in Appendix Table 1.6. None moves the green coefficient outside $[-0.0025, -0.0002]$. Appendix Table 1.6 replaces the OECD list of environmental goods with the narrower APEC list agreed in 2012; the green coefficient changes sign but not significance, confirming that the null does not depend on which products are labelled green.

Table 6: Control groups and estimation designs: EP interactions across subsamples

Sample / design	WB EP depth		TREND EP count		Obs.	Clusters
	$\times g_p$	$\times b_p$	$\times g_p$	$\times b_p$		
<i>Panel A — Non-environmental depth control (TD_{dt}, World Bank)</i>						
Full panel (reference)	−0.0023	−0.0044	−0.0001	−0.0009	21,519,511	225
bootstrap p	[0.686]	[0.185]	[0.931]	[0.177]		
Collapsed panel	−0.0046	−0.0119	+0.0018	+0.0004	3,681,023	228
bootstrap p	[0.649]	[0.072]	[0.390]	[0.857]		
C-prod-HS4	−0.0009	+0.0066	+0.0005	+0.0024	3,772,855	215
bootstrap p	[0.860]	[0.435]	[0.730]	[0.734]		
CEM-matched destinations	−0.0023	−0.0041	−0.0006	−0.0011	13,992,396	57
bootstrap p	[0.610]	[0.251]	[0.609]	[0.204]		
Deep vs. shallow (partners only)	−0.0022	−0.0034	−0.0004	−0.0006	5,262,293	23
bootstrap p	[0.585]	[0.499]	[0.751]	[0.739]		
<i>Panel B — DESTA depth control</i>						
Full panel (reference)	−0.0019	−0.0056	−0.0001	−0.0014	21,517,666	225
bootstrap p	[0.673]	[0.049]	[0.920]	[0.069]		
Collapsed panel	−0.0043	−0.0113	+0.0016	−0.0002	3,677,333	228
bootstrap p	[0.546]	[0.047]	[0.508]	[0.905]		
C-prod-HS4	−0.0009	+0.0033	+0.0004	+0.0015	3,772,321	214
bootstrap p	[0.855]	[0.850]	[0.817]	[0.735]		
CEM-matched destinations	−0.0016	−0.0043	−0.0005	−0.0013	13,992,396	57
bootstrap p	[0.676]	[0.121]	[0.676]	[0.125]		
Deep vs. shallow (partners only)	−0.0016	−0.0042	−0.0003	−0.0009	5,260,451	23
bootstrap p	[0.616]	[0.173]	[0.784]	[0.386]		

Notes: Dependent variable: log export value ($\ln x_{fpdt}^{val}$). Every cell is a separate estimation of equation (1) on the indicated sample. Fixed effects: $\theta_{fpd} + \theta_{fdt} + \theta_{pt}$ (firm-level samples), $\theta_{pd} + \theta_{dt} + \theta_{pt}$ (collapsed panel). All specifications include $TD_{dt} \times g_p$ and $TD_{dt} \times b_p$ controls (not shown). Hong Kong and Macao excluded throughout.

Inference is the wild cluster bootstrap in every cell ($B = 9,999$, clustered by destination), so all forty cells are directly comparable. Asymptotic p -values, which are unreliable with 23 treated clusters, are reported in the full tables for each subsample (Appendix Tables 1.6, 1.6 and 1.6). No stars are shown: with this many designs, star-counting invites exactly the selective reading the table is meant to prevent.

C-prod-HS4: restricted to HS6 products with a comparable HS4-level match across classification revisions. *CEM*: destinations matched on log GDP per capita, growth and MFN tariff in 2000. *Deep vs. shallow*: PTA partner destinations only, so identification comes from variation in EP depth among treated destinations rather than from treated-versus-untreated comparisons.

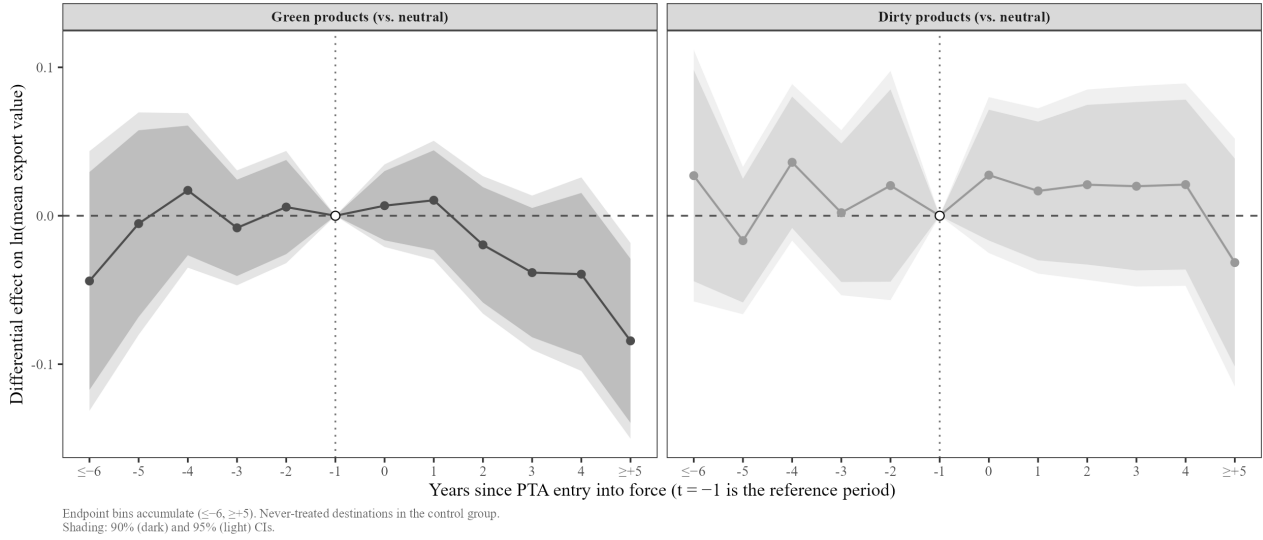
Source: Chinese Customs Office; World Bank DTA Database; TREND Database; DESTA.

1.4 Dynamics and pre-trends

Figure 1 plots the event study of the green and dirty gap relative to neutral products around entry into force, using never-treated destinations as controls. Differential pre-trends are flat on both margins.

Before reading that as reassurance, a clarification. Parallel trends is a claim about the unobserved counterfactual path of treated destinations *after* treatment. A flat pre-trend rules out the most easily detectable violation — a composition gap already in motion before the agreement — but does not establish that the assumption holds thereafter. The exercise is a falsification, not a proof.

Figure 1: Event study: green and dirty vs. neutral products around PTA entry into force



Notes: Collapsed panel; FE $pd + dt + pt$; destination-clustered 95% confidence bands; $t = -1$ is the reference period; endpoint bins accumulate. Coefficients are tabulated in Appendix Table 1.6. Source: Chinese Customs Office; World Bank DTA Database.

A mild negative drift for green products in the late post-period (bins $\geq +5$) is identified only by the early cohorts — ASEAN 2005, Chile 2006, Pakistan 2007, New Zealand 2008. It disappears under the SUN2021 interaction-weighted estimator applied to the destination-level composition gap: the aggregated ATT is -0.042 ($p = 0.27$) for green and $+0.073$ ($p = 0.28$) for dirty, consistent with early-cohort heterogeneity rather than any effect of treatment. The Sun–Abraham estimator answers a narrower question than equation (1) — it binarises treatment, discards depth, carries no depth control, and operates on a gap collapsed across products — so it is a diagnostic for staggered timing, not a replication. The full interaction-weighted event study is in Appendix Table 1.6; with standard errors that account for estimation uncertainty in the cohort-share weights, no coefficient in the $[-10, +8]$ window is individually distinguishable from zero on either margin.

Destination-specific trends. The fixed effects in equation (1) do not absorb a shock specific to green products in a particular destination that grows over time — a country signing environmental clauses precisely as its demand for green goods is rising. Two variants bracket the answer, both on the collapsed panel (Table 7).

Allowing each destination its own linear trend interacted with product type over the full sample period leaves the World Bank results essentially unchanged ($EP \times \text{green} -0.0051$, bootstrap $p = 0.49$; $EP \times \text{dirty} -0.0082$, bootstrap $p = 0.27$). The TREND green interaction, however, turns significantly negative and is the single coefficient in the entire robustness battery that survives the wild cluster bootstrap ($p = 0.015$).

This should not be taken at face value, for two reasons. First, destination-specific trends estimated over the full sample period absorb both pre- and post-treatment variation and can reverse the sign of treatment coefficients when cohort dynamics are present (WOLFERS2006); the late negative drift these trends absorb is precisely the early-cohort artefact the Sun–Abraham estimator identifies as noise. Second, and more directly, Panel B' shows the green gap was already rising before entry into force (pre-agreement slope +0.0073, bootstrap $p = 0.19$), so a pre-existing trend is a plausible account of the same coefficient.

The second variant estimates destination-specific slopes on pre-treatment years only and projects them forward before re-estimating on the detrended outcome. Every coefficient returns to an imprecise zero: TREND $\times$ green +0.0073 ($p = 0.19$), WB $\times$ green +0.0173 ($p = 0.68$), dirty interactions $p = 0.29$ –0.61. The composition null stands against destination-specific linear drifts in green demand, which is the most natural form of the selection-on-trends confounder.

Table 7: Destination-specific trends and pre-trend tests (collapsed panel)

	(1)	(2)	(3)	(4)
<i>Panel A — destination×product-type linear trends (WB index)</i>				
EP depth × Green	-0.00510 (0.00360) [0.494]	-0.00200 (0.00436) [0.808]	0.00084 (0.00434) [0.896]	-0.00006 (0.00441) [0.995]
EP depth × Dirty	-0.00825* (0.00423) [0.275]	-0.00927** (0.00414) [0.249]	-0.01030*** (0.00239) [0.086]	-0.00930*** (0.00204) [0.037]
<i>Panel A' — same specification, TREND index</i>				
EP count × Green	-0.00217*** (0.00062) [0.015]	-0.00214*** (0.00061) [0.018]	-0.00111** (0.00053) [0.218]	-0.00153*** (0.00049) [0.074]
EP count × Dirty	-0.00137 (0.00115) [0.355]	-0.00142 (0.00114) [0.339]	-0.00231*** (0.00066) [0.048]	-0.00164** (0.00064) [0.109]
<i>Panel B — pre-trend test (WB index)</i>				
Pre-agreement slope, Green	0.01732 (0.01295)	0.00402 (0.01678)	0.01719* (0.00997)	0.02218 (0.01346)
Bootstrap <i>p</i> -value	0.678	0.705	0.543	0.497
Pre-agreement slope, Dirty	0.06292 (0.04393)	0.06237 (0.03983)	0.07109** (0.03161)	0.07056** (0.03159)
Bootstrap <i>p</i> -value	0.610	0.515	0.416	0.410
<i>Panel B' — pre-trend test (TREND index)</i>				
Pre-agreement slope, Green	0.00728* (0.00438)	0.00707 (0.00441)	0.00775* (0.00428)	0.00944** (0.00389)
Bootstrap <i>p</i> -value	0.189	0.199	0.173	0.090
Pre-agreement slope, Dirty	0.01512** (0.00728)	0.01512** (0.00721)	0.01839*** (0.00596)	0.01703*** (0.00613)
Bootstrap <i>p</i> -value	0.293	0.301	0.164	0.243

Notes: Columns: (1) Excl. HK/Macao, TotalDepth control (baseline); (2) Incl. HK/Macao, TotalDepth control; (3) Excl. HK/Macao, DESTA control; (4) Incl. HK/Macao, DESTA control.

Stars are based on asymptotic *p*-values (destination-clustered standard errors). Wild cluster bootstrap *p*-values are reported separately in brackets (Panels A/A') or below each coefficient (Panels B/B').

Panels A/A'. The main specification's fixed effects do not absorb a shock specific to green products in a particular destination that grows over time — i.e., a country signing environmental clauses precisely as its demand for green goods is rising. Adding a linear trend for each destination interacted with product type absorbs exactly that dynamic.

TREND result in Panel A'. The TREND green coefficient is highly significant asymptotically ($p < 0.001$ in the baseline column) and survives the bootstrap ($p = 0.015$). Panel B' shows, however, that the green gap was already rising *before* the agreement entered into force (pre-slope +0.007, bootstrap $p = 0.189$), making a pre-existing trend a plausible explanation and weakening any causal reading.

Panels B/B'. The pre-agreement slope of the composition gap is estimated on pre-treatment years. A slope significantly different from zero would indicate that green and neutral products were already diverging before the agreement.

Standard errors in parentheses, *p*-values in brackets. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

1.5 Provision bundling and the limits of heterogeneity analysis

The indices used so far sum all environmental provisions in an agreement regardless of content. Most are cooperation clauses, institutional commitments or general environmental language, none of which has an identifiable channel through which it could shift which products a country exports. Only a small minority carry a direct trade mechanism: tariff reductions on green goods, non-regression clauses preventing countries from lowering standards to attract trade, or binding market-access commitments. A null on the aggregate index is therefore ambiguous. It could mean that environmental provisions do not affect export composition, or that the signal from a few trade-relevant clauses is diluted by a large majority that should have no effect in the first place.

Table 8 quantifies the dilution. The World Bank index checks 150 provision slots across the 25 treated destinations; 8 of them (5.3%) belong to sub-indices with a trade mechanism. The TREND database, coded at finer granularity, gives 4 of 437 (0.9%). At agreement level, a binding environmental obligation appears in exactly one of the 14–15 Chinese PTAs in the database: China–Korea, 2015.

Table 8: Trade-mechanism content as a share of total environmental provisions

	Checked	Trade-mechanism	Share
<i>Panel A: Provision-level</i>			
WB: Green Liberalization + Standards Non-Regression	150	8	5.3%
TREND: Green Market Access	437	4	0.9%
	Agreements	With clause	Share
<i>Panel B: Agreement-level</i>			
TREND: Binding environmental obligations	14–15	1	6.7%

Notes: Panel A sums each depth index across the 25 treated destinations at each destination’s maximum (post-entry) value. “Checked” is the total number of provision slots coded; “Trade-mechanism” counts those in sub-indices with a direct channel to trade flows. Panel B counts Chinese PTAs in the TREND database; the single agreement with a binding environmental obligation is the China–Korea PTA (2015). Source: World Bank DTA Database; TREND Database.

The structural consequence is stark. The two World Bank sub-indices with a direct trade mechanism are non-zero in only three destination–years, always in the same proportion, and are therefore perfectly collinear ($\rho = 1.000$) and not separately identifiable. BRANDI2020 report a maximum VIF of 4.6 across provision-type variables on a 680-agreement panel; with 13 distinct EP profiles, this design cannot do the same.

A reading rule applies to everything that follows. Each sub-index replaces the aggregate index one at a time and is estimated on its own, so the coefficients are not partial effects holding other clause types constant — they absorb whatever correlated variation moves with each one. Table 9 is a set of separate comparisons, not an additive decomposition.

Two things can be said. First, the trade-mechanism bundle reproduces the aggregate pattern: green never responds (WB Green Liberalization $\times$ green $p = 0.90$; TREND Green Market Access $\times$ green $p = 0.37$), while the dirty interaction is marginally negative on asymptotic p -values — the same asymptotics-only signal documented in Section 1.2.

Second, and more usefully, the full panel and the collapsed panel disagree on the sub-indices in ex-

actly the way they disagree on the aggregate. On the collapsed panel, WB Green Liberalization $\times$ dirty is -0.088 and TREND Green Market Access $\times$ dirty is -0.049 ; on the full panel the same coefficients are -0.002 and -0.009 and nowhere near significance (Appendix Table 1.6). The sub-index results, like the aggregate dirty coefficient, are a between-firm phenomenon that does not survive firm $\times$ destination $\times$ year fixed effects.

The placebo sub-indices deserve care. TREND soft provisions show nothing on either margin ($\times$ green $p = 0.27$; $\times$ dirty $p = 0.86$), consistent with the logic of the null. Regulatory-space clauses do not: they carry $+0.024$ on green and $+0.023$ on dirty, and both survive the wild cluster bootstrap ($p = 0.045$ and $p = 0.021$). This is the only place in the paper where a sub-index without a trade mechanism registers a signal that robust inference sustains.

Two features bound what can be read into it. The two margins move in tandem — the green-dirty differential is $+0.0017$ with $p = 0.80$ — so the coefficient describes green and dirty exports rising equally against the neutral baseline, which is not a shift in composition toward cleaner goods. And the regulatory-space sub-index is not a clean placebo: it accounts for 71.5% of the TREND count across treated destination-years and correlates 0.90 with total depth, and in this specification the depth control flips to significantly negative. Two regressors correlated at 0.90 splitting into a large positive and a large negative is the bundling problem applied to the placebo, not a cleanly identified channel.

This reconciles the null with BRANDI2020. Their finding is not that environmental provisions generically matter; it is driven by the subset of agreements with explicitly liberalizing or restrictive clauses, identified separately because 680 agreements supply enough cross-agreement variation in provision type. The Chinese PTAs of this period are, on the same coding, overwhelmingly cooperation-only. A null on the aggregate index is what their own heterogeneity would predict for a sample of this composition. A second difference compounds it: BRANDI2020 work with exporter-importer-year flows, where a composition effect can arise from the entry or exit of firms or products, whereas equation (1) identifies from reallocation within the same firm — a strictly more demanding test of the same hypothesis.

Table 9: Collapsed panel — sub-index decomposition

	WB sub-indices		TREND sub-indices				
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
	Green Lib.	Enf. DSM	Green Mkt.	Enf. DSM	Hard	Soft	Reg. Space
<i>Panel A — Non-environmental depth control (TD_{dt}, World Bank)</i>							
$EP_{\text{sub}} \times g_p$	-0.0115 (0.0879)	-0.0048 (0.0394)	-0.0271 (0.0303)	0.0042 (0.0148)	0.0022 (0.0047)	0.0132 (0.0121)	0.0242** (0.0099)
$EP_{\text{sub}} \times b_p$	-0.0876* (0.0483)	0.0067 (0.0525)	-0.0491** (0.0243)	-0.0052 (0.0140)	-0.0013 (0.0037)	0.0019 (0.0108)	0.0225*** (0.0086)
θ_{pd}	✓	✓	✓	✓	✓	✓	✓
θ_{dt}	✓	✓	✓	✓	✓	✓	✓
θ_{pt}	✓	✓	✓	✓	✓	✓	✓
Observations	3,681,023	3,681,023	3,681,023	3,681,023	3,681,023	3,681,023	3,681,023
Clusters (d)	228	228	228	228	228	228	228
<i>Panel B — DESTA depth control</i>							
$EP_{\text{sub}} \times g_p$	-0.0141 (0.0892)	-0.0105 (0.0280)	-0.0290 (0.0288)	0.0032 (0.0160)	0.0017 (0.0047)	0.0132 (0.0131)	0.0149 (0.0104)
$EP_{\text{sub}} \times b_p$	-0.1001** (0.0398)	-0.0197 (0.0343)	-0.0543*** (0.0187)	-0.0086 (0.0115)	-0.0023 (0.0029)	-0.0015 (0.0101)	0.0092 (0.0075)
θ_{pd}	✓	✓	✓	✓	✓	✓	✓
θ_{dt}	✓	✓	✓	✓	✓	✓	✓
θ_{pt}	✓	✓	✓	✓	✓	✓	✓
Observations	3,677,333	3,677,333	3,677,333	3,677,333	3,677,333	3,677,333	3,677,333
Clusters (d)	228	228	228	228	228	228	228

Notes: Collapsed panel (HS6–destination–year cells), excluding Hong Kong and Macao. Dependent variable: mean log export value ($\bar{y}_{pdt}$), weighted by cell size. Each column estimates a separate model replacing the aggregate EP index with one sub-index; the non-environmental depth control ($TD_{dt} \times g_p$ and $TD_{dt} \times b_p$) is included in all specifications but not shown. Fixed effects: product-destination (θ_{pd}), destination-year (θ_{dt}), product-year (θ_{pt}). Standard errors clustered at the destination level in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

WB sub-indices: (1) Green Liberalization; (2) Enforcement & DSM. TREND sub-indices: (3) Green Market Access; (4) Enforcement & DSM; (5) Hard obligations; (6) Soft obligations; (7) Regulatory Space.

1.6 The extensive margin

Log-linear estimates condition on positive flows. If environmental provisions create new green trade where none previously existed, the estimates above miss it (SANTOSSILVA2006). On an HS6–destination–year grid completed with zeros (8.2 million cells, of which 7.9 million enter estimation after PPML drops separated and singleton observations), the picture is unchanged (Table 10): $EP \times \text{green}$ is $+0.0015$ ($p = 0.74$, WB) and $+0.0001$ ($p = 0.95$, TREND), and $EP \times \text{dirty}$ is -0.030 ($p = 0.057$, WB) and $+0.003$ ($p = 0.55$, TREND). No green trade creation is detected at the extensive margin, and the dirty coefficient is at most marginal, consistent with Section 1.2.

One limitation should be noted. This grid detects a green product newly exported to a destination, but not a new firm beginning to export a product that other Chinese firms already ship there, which is a common form of firm-level entry. The extensive margin is probed, not exhausted.

Table 10: PPML — extensive margin (collapsed panel, zero-filled)

	WB EP depth (EP_{dt}^{WB})		TREND EP count (EP_{dt}^{TR})	
	(1)	(2)	(3)	(4)
<i>Panel A — Non-environmental depth control (TD_{dt}, World Bank)</i>				
$EP_{dt} \times g_p$	0.0015 (0.0046)	0.0029 (0.0061)	0.0001 (0.0019)	−0.0001 (0.0021)
Asymptotic p	[0.738]	[0.630]	[0.951]	[0.972]
$EP_{dt} \times b_p$	−0.0301* (0.0158)	−0.0268 (0.0178)	0.0028 (0.0047)	0.0048 (0.0050)
Asymptotic p	[0.057]	[0.133]	[0.550]	[0.335]
θ_{pd}	✓	✓	✓	✓
θ_{dt}	✓	✓	✓	✓
θ_{pt}	✓	✓	✓	✓
Observations	7,895,543	8,030,137	7,895,543	8,030,137
Clusters (d)	225	228	225	228
<i>Panel B — DESTA depth control</i>				
$EP_{dt} \times g_p$	0.0053 (0.0045)	0.0064 (0.0050)	0.0010 (0.0013)	0.0008 (0.0015)
Asymptotic p	[0.232]	[0.205]	[0.442]	[0.565]
$EP_{dt} \times b_p$	−0.0186 (0.0169)	−0.0108 (0.0198)	0.0027 (0.0053)	0.0046 (0.0050)
Asymptotic p	[0.270]	[0.585]	[0.612]	[0.358]
θ_{pd}	✓	✓	✓	✓
θ_{dt}	✓	✓	✓	✓
θ_{pt}	✓	✓	✓	✓
Observations	7,878,784	8,003,772	7,878,784	8,003,772
Clusters (d)	225	228	225	228

Notes: PPML estimated on the HS6–destination–year grid completed with zeros (Santos Silva & Tenreyro, 2006); fixed effects: product×destination (θ_{pd}), destination×year (θ_{dt}), product×year (θ_{pt}); standard errors clustered at the destination level. Columns (1) and (3) exclude Hong Kong and Macao; columns (2) and (4) include them. Panel A controls for non-environmental WB depth ($TD_{dt} \times g_p$ and $TD_{dt} \times b_p$, not shown); Panel B uses the DESTA depth index.

Log-linear estimates condition on positive flows: if an environmental clause created new green exports to a market where none previously existed, those estimates would miss it. PPML retains zeros and therefore captures this margin.

PPML coefficients are approximate percentage changes.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$ (asymptotic).

Note on what moved to the appendix

Saturation ladder (with a baseline column to be added, so the sequence terminates at Table 1); Sun–Abraham event study; tabulated TWFE event study; Hong Kong and Macao variants of both main tables; full wild-cluster-bootstrap and permutation tables; full-panel robustness battery; APEC versus OECD product definition; CO₂ intensity as a continuous dirty measure; sub-index estimates on the full panel; minimum detectable effects; environmental share of agreement content (ABMAN2024); the full subsample tables underlying Table 6.

Open drafting decisions. (i) Table 7 is included whole; Panels A/A' could move to the appendix leaving only the pre-trend test in the main text. (ii) The saturation-ladder sign reversal on the dirty margin (+0.0064 under $fpd + pt$, −0.0044 in the reference specification) is a strong fragility argument and currently sits only in the appendix; it could be a paragraph in Section 1.2. (iii) Table 1.6 is estimated on the collapsed panel while the bounding paragraph in Section 1.1 uses full-panel intervals; these should be harmonised.

Appendix placeholders (labels only, tables live in the table document)

- Saturation ladder
- Sun–Abraham event study
- Event study (TWFE), tabulated
- Full panel incl. HK/Macao
- Collapsed panel incl. HK/Macao
- Wild cluster bootstrap, all variants
- Permutation test, all variants
- Full-panel robustness battery
- APEC vs. OECD green definition
- CO₂ intensity (Shapiro)
- Sub-indices, full panel
- Minimum detectable effects
- C-prod-HS4, full table
- CEM destinations, full table
- Deep vs. shallow, full table